

StateLens: Linear Attention — One-Page Concept Summary

DataForge 2026, Pathway Track | StateLens (github.com/yogeshroyal63-beep/Dataforge) | ~720 words

1. The design pressure

Standard scaled dot-product attention (Vaswani et al., 2017) answers each query by comparing it against every past key–value pair — an $O(N)$ memory footprint and $O(N)$ per-query compute cost that grows with sequence length. For long documents or many in-context demonstrations, retaining the full key–value cache becomes the primary bottleneck: GPU memory, inference latency, and deployment cost all scale linearly with context length. The design pressure is real and actively motivating a wave of Post-Transformer architectures (Mamba, RetNet, GLA, BDH) that attempt to answer queries in constant memory.

2. What linear attention changes

By replacing softmax with a positive feature map ϕ , the attention equation factors so the sum over past keys and values can be accumulated incrementally as a running outer product (Katharopoulos et al., 2020, ICML). The recurrence is:

$$\begin{aligned} S_t &= 0 & z_t &= 0 & S_t &= S_{t-1} + \phi(k_t) v_t & z_t &= z_{t-1} + \phi(k_t) & y_t &= \phi(q_t) S_t / \phi(q_t) z_t \\ z_t \end{aligned}$$

S_t is a fixed-size $d_k \times d_v$ matrix — it never grows with sequence length. Every new token is folded additively into the same memory slots. This reduces memory footprint from $O(N)$ to $O(1)$ and per-query compute from $O(N)$ to $O(1)$.

3. The one-sentence claim StateLens teaches

"A linear-attention model can track information across a sequence using a fixed-size running state instead of storing every past token, but as more facts are packed into that fixed state, older facts increasingly interfere with and get overwritten by newer ones — a failure mode standard attention does not have."

4. Representative architectures

System	State size	Update rule	Key trade-off
Full Attention (Vaswani 2017)	$O(N)$ KV cache	Append token	Perfect recall; $O(N)$ memory
Linear Attn (Katharopoulos 2020)	Fixed $d \times d$ matrix	Additive outer product	$O(1)$ memory; state interference
GLA (Yang 2023)	Fixed $d \times d$ + gate	Gated decay $S_t = G_t S_{t-1} + \dots$	Softer forgetting; gate overhead
RetNet (Sun 2023)	Fixed $d \times d$ + decay	Exponential decay	Stable training; fixed forget rate
BDH-CQ (Engdahl 2026)	High-dim latent workspace	Recurrent demo accumulation	No CoT; $O(1)$ inference state

5. Evidence and what it establishes

StateLens toy benchmark (live, client-side): The app runs `runFullBenchmark()` — associative recall at $N = 2 \dots 20$ facts, 5 seeds per condition — entirely in the browser. At $N = 20$ (VOCAB cap), full attention achieves near-perfect recall; linear attention drops to roughly 34% on average. This is the interference effect the claim predicts. These numbers are computed by the visitor's own browser and are reproducible from source; they are not a published research result.

BDH-CQ published result (Engdahl et al., 2026, arXiv:2608.09888): A 150M-parameter BDH-CQ model achieved 29.5% pass@2 on ARC-AGI-1 at \$0.00070 per task (~0.85 s of H200 time). The authors report this operating point as lying on the cost–accuracy Pareto frontier — more accurate per dollar than comparable systems at the time. This figure is Pathway's own published benchmark result, not independently reproduced by StateLens.

6. Role of BDH and BDH-CQ

BDH (Dragon Hatchling; Kosowski et al., 2025, arXiv:2509.26507) is a brain-inspired Post-Transformer architecture where attention is reformulated as Hebbian synaptic memory: connection strengths between neuron units update as the model

reads, implementing a form of working memory without a separate cache. The GPU-friendly BDH-GPU formulation uses ReLU-low-rank transformations with linear attention.

BDH-CQ extends this to in-context learning: demonstrations of a new task are accumulated into a recurrent latent memory state without any weight update. The BDH-CQ technical report explicitly frames this contextual memory using fast-weight and linear-attention language — the same additive fixed-state accumulation StateLens demonstrates in Part A. The key parallel BDH-CQ's own paper draws: $S_t = S_{t-1} + \phi(k_t) v_t$, where each demonstration updates the same high-dimensional recurrent workspace.

7. Limitations and open questions

StateLens is a toy model. KEY_DIM = 16, VALUE_DIM = 10, VOCAB cap = 20 words. The interference pattern is structurally correct but the toy numbers are not comparable to any published result at scale.

BDH-CQ's own documented failure: The technical report records complete failure (0 of 72 tasks) on a color-swap composition probe, suggesting an object-property binding limitation under distribution shift.

Evidence classification: BDH-CQ's 29.5% ARC-AGI-1 figure is a developer benchmark, not an independent evaluation. Some architectural and training details remain undisclosed, limiting external reproducibility.

8. Where to continue

Katharopoulos et al. (2020) (arXiv:2006.16236): Foundational linear-attention recurrence.

Yang et al. (2023) — GLA (arXiv:2312.06635): Gated-decay fix for interference.

Sun et al. (2023) — RetNet (arXiv:2307.08621): Fixed-decay variant, large-scale evals.

Kosowski et al. (2025) — BDH (arXiv:2509.26507): Dragon Hatchling paper.

Engdahl et al. (2026) — BDH-CQ (arXiv:2608.09888): BDH-CQ technical report.